ModelIt!

Teacher's Guide

Why Are Coral Reefs Dying?

Modeling Ocean Acidification, Thermal Bleaching, and Ecosystem Collapse

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-6

Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations for why coral reefs are dying based on their models of symbiosis breakdown, acidification, and phase shifts, and design multi-scale intervention strategies.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — A complex set of interactions within an ecosystem can keep its numbers and types of organisms relatively constant over long periods of time under stable conditions. If a modest biological or physical disturbance to an ecosystem occurs, it may return to its more or less original status, but extreme changes may result in a very different ecosystem.
Crosscutting Concept	**Stability and Change** — Students analyze how coral reef ecosystems can absorb moderate disturbances and return to stable states, but extreme or repeated disturbances can push the system past a tipping point into a fundamentally different and degraded state.

CAST Testing Alignment

MULTIPLE CHOICE:

A coral reef has experienced three bleaching events in eight years. According to your model, why is the third event more damaging than the first even if water temperatures are similar? A) The coral has adapted to be more heat-sensitive B) Ocean acidification has worsened C) Incomplete recovery left coral weakened and algae encroachment reduced available recovery space D) Bleaching always gets worse over time regardless of conditions

CONSTRUCTED RESPONSE:

Using your reef ecosystem model, explain why protecting herbivorous fish populations from overfishing could significantly increase a coral reef\'s resilience to climate change. Reference at least three components and one feedback loop in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Coral reefs are among the most biodiverse and economically valuable ecosystems on Earth, supporting approximately 25% of all marine species while covering less than 1% of the ocean floor. These underwater rainforests are built by tiny coral polyps — animals related to jellyfish — that form a remarkable symbiosis with photosynthetic algae called zooxanthellae. This partnership has built massive limestone structures over thousands of years, but it is extraordinarily sensitive to changes in water temperature and chemistry that are now occurring at unprecedented rates.

The Coral-Zooxanthellae Symbiosis

Coral polyps are animals that cannot photosynthesize, yet they thrive in nutrient-poor tropical waters because of their partnership with zooxanthellae — single-celled algae that live within the coral\'s tissue. The zooxanthellae photosynthesize using sunlight and provide up to 90% of the coral\'s energy needs in the form of sugars and amino acids. In return, the coral provides the algae with shelter, CO2, and nutrients from its waste. This symbiosis is the engine that drives reef productivity, enabling corals to build their calcium carbonate skeletons and support the extraordinary biodiversity that makes reefs the rainforests of the sea. But this partnership has a fatal vulnerability: it breaks down when water temperature exceeds the coral\'s tolerance threshold by even 1-2 degrees Celsius.

Bleaching: When Symbiosis Breaks Down

When sea surface temperature exceeds the coral\'s thermal tolerance — typically 1-2 degrees above the normal summer maximum — the zooxanthellae begin producing toxic reactive oxygen species instead of beneficial sugars. The coral responds by expelling its algal partners, revealing the white calcium carbonate skeleton beneath — this is bleaching. A bleached coral is not dead but is starving, surviving on stored energy reserves for 4-8 weeks. If temperatures return to normal within this window, the coral can reacquire zooxanthellae and recover, though it remains weakened for months. If high temperatures persist or return too quickly, the coral dies. The critical problem is frequency: the Great Barrier Reef experienced mass bleaching in 1998, 2002, 2016, 2017, 2020, 2022, and 2024 — recovery requires 10-15 years of stable conditions that no longer exist.

Ocean Acidification: The Other CO2 Problem

While atmospheric CO2 drives warming through the greenhouse effect, dissolved CO2 directly alters ocean chemistry. When CO2 dissolves in seawater, it forms carbonic acid, releasing hydrogen ions that lower pH and binding with carbonate ions to form bicarbonate. This reduces the concentration of carbonate ions that corals, shellfish, and many marine organisms need to build calcium carbonate structures. Ocean pH has already dropped from 8.2 to 8.1 since pre-industrial times — a 26% increase in acidity that sounds small but is logarithmic. Under current emission trajectories, ocean pH could drop to 7.8 by 2100, representing a 150% increase in acidity from pre-industrial levels. At this pH, coral reefs dissolve faster than they can grow, and many marine organisms cannot form shells or skeletons.

Phase Shifts: The Point of No Return

When coral cover drops below a critical threshold — typically around 10-20% — the reef undergoes a phase shift to an algae-dominated state. This shift involves multiple positive feedback loops: dead coral provides substrate for algae, algae smother coral larvae preventing recolonization, reduced coral structure drives away herbivorous fish that would control algae, and the algae-dominated reef supports fewer organisms that maintain reef health. Once established, these feedback loops make the phase shift extremely difficult to reverse — removing one stressor is insufficient because the others maintain the degraded state. Studies of Caribbean reefs show that many crossed this threshold in the 1980s following disease that killed sea urchins (key herbivores), and 40 years later most have not recovered despite management efforts.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Sea Surface Temperature, Ocean pH Level, Coral-Zooxanthellae Symbiosis, Coral Calcification Rate, Reef Biodiversity Index, Algae Overgrowth, and Herbivore Fish Population as key components of the reef ecosystem.

E **Establish Relationships**

Students determine that temperature and pH stress the coral-algae symbiosis, that symbiosis breakdown reduces calcification and biodiversity, that algae overgrowth creates positive feedback loops preventing recovery, and that herbivore fish populations moderate algae growth.

V **Visualize & Model**

Students build a computational model showing how ocean conditions affect coral health through multiple interacting pathways and identify the feedback loops that can lock reefs into degraded states.

E **Evaluate Outcomes**

Students run moderate warming, double-stress, and managed reef scenarios to test predictions about bleaching recovery, phase shift thresholds, and the effectiveness of local management interventions.

R **Revise & Extend**

Students add nutrient pollution, coral genetic diversity, or marine protected area coverage to explore how additional factors affect reef resilience and recovery potential.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Today we\'re investigating one of the most urgent ecological crises happening right now — the death of the world\'s coral reefs. Why are they dying, and can we save them?"*

Watch For

Show cover image. Quick share: Has anyone ever seen a coral reef — in person, in a documentary, or online? What did it look like?

Slide 2: Learning Objectives

3 min

Say: *"By the end of today, you\'ll understand the science behind coral bleaching, ocean acidification, and why reefs can collapse permanently once they pass a tipping point."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key concept: zooxanthellae — these microscopic algae are the key to everything. Use the symbiosis analogy: coral provides housing, algae pays rent in sugar.

Slide 3: Big Question

3 min

Say: *"Coral reefs support a quarter of all ocean life on less than 1% of the seafloor. What happens to that quarter when the reefs die? Write your prediction."*

Watch For

Quick-write: Students write their initial prediction about cascading effects of reef loss. Save for comparison after modeling.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll build a model of the coral reef ecosystem to understand not just what\'s happening, but why reefs can\'t recover once they pass certain thresholds."*

Watch For

Preview each LEVER step. Emphasize that this model will reveal feedback loops — processes that make reef death self-reinforcing once they start.

Slide 5: Activity 1: Components

8 min

Say: *"Which of these components are external drivers that come from outside the reef system, and which are internal responses within the reef ecosystem?"*

Watch For

Guide sorting: External = Sea Surface Temperature, Ocean pH Level (driven by global processes). Internal = the five ecosystem responses. Discuss: Why is Herbivore Fish Population internal even though fishing is an external pressure?

Slide 6: Activity 2: Connections

8 min

Say: *"When temperature rises and triggers bleaching, trace the cascade through the system. What happens to the symbiosis, then to calcification, then to biodiversity, then to algae? Look for loops."*

Watch For

Students draw arrows with + or - signs. Key discovery: the positive feedback loop between Algae Overgrowth, Coral-Zooxanthellae Symbiosis, and Herbivore Fish Population. Ask: Once this loop starts, can it stop itself?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios and watch what happens. Pay special attention to the managed reef scenario — it reveals something important about what we can actually do."*

Watch For

Students run each scenario, record predictions vs. observations. Key discussion: Compare the double-stress scenario to the managed scenario. What does this tell us about the value of local conservation?

Slide 8: Discoveries

5 min

Say: *"Our model reveals that reef death is not just about heat — it\'s about feedback loops, recovery time, and the interaction of multiple stressors. Let\'s discuss what this means for conservation."*

Watch For

Class discussion of discoveries. Compare initial predictions to model evidence. Key question: What surprised you about the managed reef scenario?

Slide 9: STEM Challenge

12 min

Say: *"Your reef is in crisis but not yet past the point of no return. Use your model data to design a rescue plan that combines local management with long-term strategy."*

Watch For

Groups design multi-scale intervention plans. Must address both local stressors (fishing, pollution) and global stressors (warming, acidification). Present strategies with evidence from model simulations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sea Surface Temperature, Ocean pH Level
- **INTERNAL:** Coral-Zooxanthellae Symbiosis, Coral Calcification Rate, Reef Biodiversity Index, Algae Overgrowth, Herbivore Fish Population

Relationships

- **Sea Surface Temperature to Coral-Zooxanthellae Symbiosis:** NEGATIVE (-) — Rising temperature above the thermal tolerance triggers bleaching — corals expel their zooxanthellae algae, weakening or destroying the symbiosis that provides 90% of the coral\'s energy.
- **Ocean pH Level to Coral Calcification Rate:** POSITIVE (+) — As pH decreases (more acidic), carbonate ion concentration drops, directly slowing the rate at which corals can build and maintain their calcium carbonate skeletons. Lower pH means weaker, slower-growing reefs.
- **Coral-Zooxanthellae Symbiosis to Reef Biodiversity Index:** POSITIVE (+) — Healthy symbiosis drives coral growth and reef structure complexity, which provides habitat for thousands of species. When symbiosis breaks down and coral dies, the structural foundation of biodiversity disappears.
- **Algae Overgrowth to Coral-Zooxanthellae Symbiosis:** NEGATIVE (-) — Algae physically smother coral and block light needed by zooxanthellae for photosynthesis. Algae overgrowth on dead coral also prevents coral larvae from settling and establishing new symbioses — a positive feedback loop.
- **Herbivore Fish Population to Algae Overgrowth:** NEGATIVE (-) — Parrotfish, surgeonfish, and other herbivores graze on algae, keeping it in check and maintaining open substrate for coral recruitment. Higher herbivore populations mean less algae overgrowth.

Key Discoveries

- Coral bleaching becomes lethal not from a single event but from repeated events that prevent full recovery — the model shows that recovery requires 10-15 years of stable conditions, but bleaching events are now occurring every 2-3 years
- Ocean acidification and warming create a compounding double stress — acidification weakens coral skeletons while warming disrupts the symbiosis, and the combined effect is worse than either stressor alone
- Phase shifts from coral to algae dominance involve positive feedback loops — dead coral provides substrate for algae, algae smother surviving coral larvae, and reduced reef structure drives away herbivorous fish that would otherwise control algae
- Maintaining healthy herbivore fish populations significantly increases reef resilience by controlling algae overgrowth and giving coral time to recover between stress events — local management can buy time even as global stressors increase

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Coral is a plant or a rock, not a living animal"	Corals are colonies of tiny animals called polyps, related to jellyfish and sea anemones. Each polyp has a mouth, tentacles, and a stomach. They build calcium carbonate skeletons over time, creating the rock-like reef structure. The color comes from their symbiotic zooxanthellae algae, not from the coral animal itself. Understanding that coral is alive — and as vulnerable as any animal — is essential to understanding bleaching and death.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bleaching means the coral is dead"	Bleaching is a stress response, not death — it means the coral has expelled its zooxanthellae but is still alive, surviving on energy reserves for 4-8 weeks. If temperatures return to normal quickly, the coral can reacquire algae and recover, though it will be weakened for months. Bleaching becomes fatal only when high temperatures persist or when events are so frequent that coral cannot rebuild between them. The distinction matters because it means we have a window of opportunity.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Coral reefs only matter to tropical countries"	Reef loss affects the entire global food web and economy. Reefs provide nursery habitat for commercially important fish species that migrate globally. Reef-derived compounds are used in medicines treating cancer, HIV, and cardiovascular disease. Reefs buffer coastlines from storms, protecting billions of dollars in infrastructure. The global economic value of reef services is estimated at $375 billion per year. Furthermore, reef collapse can trigger cascading effects in ocean food webs that affect fisheries worldwide, including in temperate and polar regions.